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Autodesk® Scaleform® 4.4

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- ...

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1

Scaleform® Common Lightweight Interface Kit
CLIK™) CLIK CLIK¼
CLIK Scaleform
CLIK CLIK

1.1

Scaleform CLIK ActionScript™ 2.0 (AS2) Scaleform
4.0 PC
CPU UI
CLIK 15

1.1.1. X Scaleform CLIK

UI:

Button	Slider
ButtonBar	StatusIndicator
CheckBox	ProgressBar
RadioButton	UILoader
Label	ScrollingList
TextInput	TileList
TextArea	DropdownMenu
ScrollIndicator	Dialog
ScrollBar	NumericStepper

API-

InputManager	Locale
FocusManager	Tween
DragManager	DataProvider
PopUpManager	IDataProvider
EventDispatcher	IList
Delegate	IListItemRenderer

Constraints	GameDataProvider

CLIK
CLIK
CLIK API
CLIK
CLIK Flash
CLIK

1.2

Flash Flash

Scaleform

CLIK Scaleform gskinner.com

gskinner.com Grant Skinner Adobe Flash Creative Suite® 4 (CS4) Flash gskinner.com
http://gskinner.com/blog

CLIK Flash studio CLIK
C:\Program Files\Scaleform\GFx 4.4\Resources\AS2\CLIK\components\CLIK_Components.fla

1.3 UI

UI Microsoft Visio® Flash UI

Scaleform CLIK

UI Flash Visio
Flash

UI

web

Flash

Flash

UI Flash

U

I UI

Scaleform Flash ActionScript UI

Lua UnrealScript

Flash ActionScript Flash

2

È Scaleform

CLIK UI CLIK

" " CLIK

CLIK UI

- ***User interaction*** :
- ***Component setup*** Flash
- ***States***
- ***Inspectable properties*** Flash
- ***Events*** UI
- ***Tips and tricks***

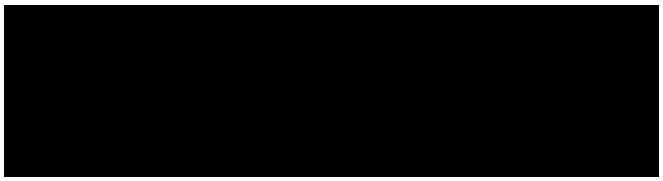
2.1

Button CheckBox Label TextInput
TextArea Button CheckBox Button Label
TextInput TextArea.

. 1 *Free Realms*.

. 2 *Free Realms* Check box

. 3 *Free Realms* Label



. 8 *Crysis Warhead* Text input

5 *Mercenaries 2* Text area

2.1.1 Button

. 6

Button CLIK Butt
on gfx.controls.Button textField
ScrollBar Slider

. 7Ö AnimatedButtonÃ AnimatedToggleButton¼ ToggleButton

CLIK
Button
ToggleButton AnimatedButton
AnimatedToggleButton Button.fla

2.1.1.1

Programming
CLIK
(Tab)
¼(Shift+Tab)
CLIK CLIK Flash
CLIK CLIK (Tab)
¼(Shift+Tab)

Enter

Enter
Xbox360
PS3 A X UI CLIK

2.1.1.2

CLIK Button MovieClip

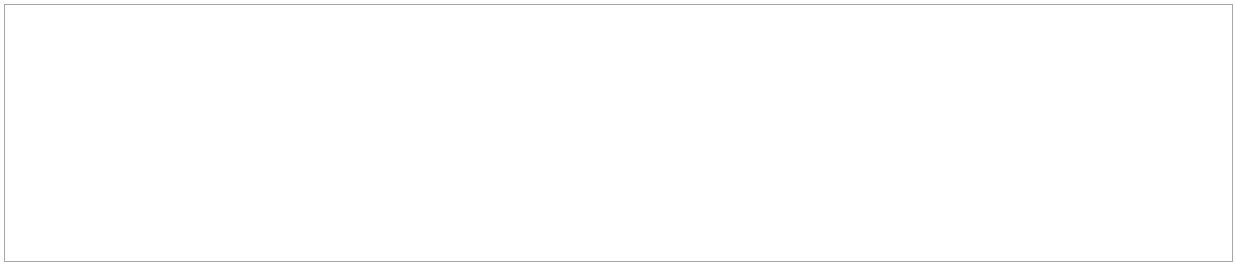
- ***textField***: ( ) TextField
- ***focusIndicator:*** ( ) MovieClip. MovieClip
  hide over
  Button
  over

2.1.1.3

CLIK

- ***up***

- ***over***
- ***down***
- ***disabled***



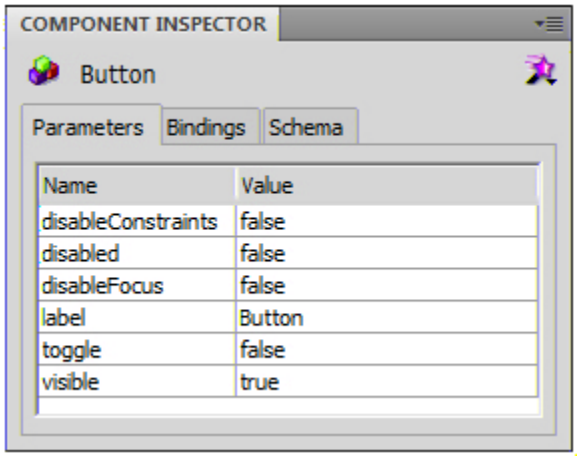
. 8Ö Button

Flash

Getting Started with CLIK Buttons

2.1.1.4

Flash IDE, Component Inspector Parameters CS4 Window Component Inspector ; (Shift+F7) Component Inspector AS



. 9Ö CS4

j :

disableConstraints	Button textField

	true textField disabled textField textField /
Disabled	true disabled
disableFocus	ture
Label	
Toggle	true
Visible	false

SWF Flash
IDE CLIK

2.1.1.5 &

CLIK

- ***type***: &Ä
- ***target:***

Show	true
Hide	false
focusIn	
focusOut	
Select	*Selected* true
stateChange	*State* String2 upÃ overÃ down1 CLIK
rollOver	*mouseIndex* -

		0-3		
rollOut				
	mouseIndex			
		0-3		
Press				
	mouseIndex			-
		0-3		
doubleClick		*doubleClickEnabled*	true	
	mouseIndex			-
		0-3		
Click				
	mouseIndex			-
		0-3		
dragOver				
	mouseIndex			-
		0-3		
dragOut				
	mouseIndex			-
		0-3		
releaseOutside				
	mouseIndex			-
		0-3		

ActionScript

```
myButton.addEventListener("click", this, "onButtonPress");
function onButtonPress(event:Object) {
      // □|□
}
```

1□>|□.1g□ click□□□-□□=0 g□ myButton□□□□3+□ onButtonPress函数。相同的代码类型也可以用来处理其他的事件。事件句柄中返回的Object参数（例子中名为‘event’）包含了事件的相关信息。这些信息作为Object参数属性返回并与多数事件有所不同。

2.1.1.6 ¼

[illegible]

```
import gfx.controls.Button;
attachMovie("Button", "btnInstanceName", someDepth, {_x:33, _y:262, …});
```

CLIK Toggle

```
btn.toggle = true;
```

□□□

- ***textField***: (ÿ) TextField 2è®
- ***focusIndicator:*** (ÿ) MovieClip 2è
  MovieClipæ MovieClip show¼ hideÆ

2.1.2.3 (e

ž Toggleè CheckBoxè

- ***up*** (æ
- ***over*** (æ
- ***down*** (æ
- ***disabled*** (æ
- ***selected_up*** (æ
- ***selected_over*** (æ
- ***selected_ down*** (æ
- ***selected_disabled*** (æ

. **11: CheckBox æ**

CheckBoxè Buttonæè
æ CLIK

2.1.2.4 è

CheckBoxè è disableFocusè Toggl
eè

data	Check Boxæ
disableConstraints	textFieldæ true textFieldæ disabled textFieldæ ...textFieldæ
disabled	true

label		
selected	true	CheckBox
visible	false	

2.1.2.5 &

Object

- **type:** 2
- **target:**

CheckBox

show		true
hide		false
focusIn		
focusOut		
select	*Selected*	true
stateChange	*State* String2 up over down1 CLIK	
rollOver	*mouseIndex* 1 2 0-3	
rollOut	*mouseIndex* 2 0-3	
press	*mouseIndex* 1 2 0-3	
doubleClick	*doubleClickEnabled* *mouseIndex* 1 2 0-3	
click	*mouseIndex* 1 2 0-3	
dragOver		

mouseIndex	
	0-3

dragOut

Label Label
textField Label textFiel
d Label

2.1.3.1

Label

2.1.3.2

CLIK Label MovieClip

- ***textField***: TextField Label

2.1.3.3

CLIK Label

- ***default*** enabled
- ***disabled***

. 13: Label

2.1.3.4

Label

text		
visible		false
disabled		true

2.1.3.5

Object

- ***type***:

- ***target***:

 Label

show	F	true
hide		false
stateChange	Label	
	State Label	String2 default F disabled Ä

 Label

```
myLabel.addEventListener("stateChange", this, "onStateChange");
function onStateChange(event:Object) {
	if (event.state == "default") {
		// 
	}
}
```

2.1.3.6

 Label HTML textField HTML Flash 8

```
lbl.htmlText = "<b>foo</b>bar";
```

2.1.4 TextInput

. 14: TextInput.

 TextInput

(nfu.controls.TextInput) textField. Label

 textField textField HTML

 TextInput, textField

textField Label Label
textFields TextInput UI PC
TextInput textField

TextInput disabled textField
TextInput TextInput AS2 TextInput
TextInput TextInput Flash
Studio TextInput

TextInput textField, HTML
(Shift+Arrows) (Shift+Delete)
(Ctrl+Insert) (Shift+Insert)

actAsButton
over out
actAsButton "over"/"out"

(0xAAAAAA)

2.1.4.1

TextInput textField

textField

2.1.4.2

CLIK TextInput MovieClip Ö

- ***textField***: textField 2

2.1.4.3 States

CLIK TextInput disabled

- ***default*** F enabled
- ***focused*** textField
- ***disabled***

. 15: TextInput

2.1.4.4

TextInput

text	textField
visible	false
disabled	true
editable	false TextInput
maxChars	textField
password	true textField '*'
defaultText	
actAsButton	tab

2.1.4.5

Object

- ***type***:
- ***target***:

TextInput组件产生的事件列表如下。事件旁边的属性列表为通用属性的补充内容。

show		true
hide		false
focusIn		
focusOut		
textChange	textField	
rollover	*mouseIndex*:	0 3.
rollout	*mouseIndex*:	0 3.

textField

```
myTextInput.addEventListener("textChange", this, "onTextChange");
function onTextChange(event:Object) {
      // ⍁|□
}
```

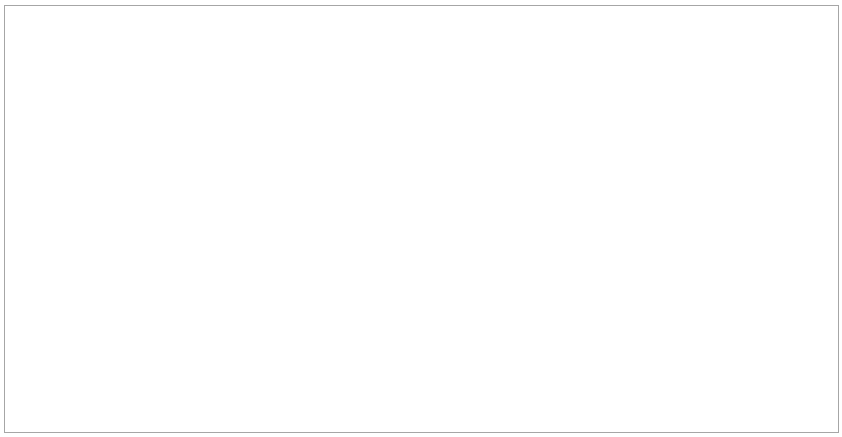
2.1.4.6 ¼

```
_global.gfxExtensions = true;
textinput.textField.noAutoSelection = true;
```

在TextInput组件中显示HTML文本　关于在标准textField 中支持　　标签请参考　　文档：

```
textinput.htmlText = "<b>foo</b>bar";
```

2.1.5 TextArea



. 16: þ TextArea.

j TextArea (gfx.controls.TextArea)þ CLIK
TextInput ScrollBar
ã

Label¼TextInputÈ TextAreaþ textFieldÈ textFieldþ/
È HTML
textField/
Area

ñh textFieldò ScrollIndicatorõ ScrollBarÈ TextAreaþ/
ô textFieldÈ TextAreaþ
TextAreaþ.

2.1.5.1 þ

& TextAreaþ textFieldþ
textFieldþ
þ

2.1.5.2 A

CLIK TextAreaþ MovieClipþ

- ***textField***: TextField 2

2.1.5.3 (e

& TextInput TextArea disabled

- ***default*** F enabled (
- ***focused*** (textField
- ***disabled*** (

17: TextArea

2.1.5.4

TextArea TextInput CLIK
ScrollBar 2.4

text	textField\
visible	false
disabled	false
editable	false TextInput
maxChars	textField
scrollBar	CLIK ScrollBar ScrollBar ID TextArea
scrollPolicy	auto on ScrollBar off ScrollBar ScrollBar
defaultText	
actAsButton	tab

2.1.5.5 &

Object

- ***type:***
- ***target:***

TextArea

show	&	"	true
hide	&	"	false
focusIn	&		
focusOut			
textChange	textField		
scroll			
rollover	*mouseIndex*:		0` 3.
rollout	*mouseIndex*:		0` 3.

. TextArea

```
myTextArea.addEventListener("scroll", this, "onTextScroll");
function onTextScroll(event:Object) {
      // |
}
```

2.1.5.6

```
_global.gfxExtensions = true;
textinput.textField.noAutoSelection = true;
```

TextArea/j HTML textField HTML Flash 8

```
textinput.htmlText = "<b>foo</b>bar";
```

2.2

RadioButton ButtonGroup

ButtonBar ButtonGroup Button

ButtonBar RadioButton

ButtonGroup

18: *Dawn of War II*

2.2.1 RadioButton

. 19: **RadioButton**

RadioButton

(javafx.scene.control.RadioButton)

CLIK CheckBox

ButtonGroup

2.2.1.1 用户交互

2.2.1.2 组件配置

使用CLIK RadioButton 的MovieClip必 具 下面所列的子 元。也列出了可 元素：

- ***textField***（可 ）TextField 实例。
- ***focusIndicator***：MovieClip MovieClip MovieClip hide

2.2.1.3 状态

CheckBox

- ***up*** 状态
- ***over*** 状态
- ***down*** 状态
- ***disabled*** 状态
- ***selected_up*** 状态
- ***selected_over*** 状态
- ***selected_ down*** 状态
- ***selected_disabled*** 状态

. 20:5 RadioButton

5 RadioButton Button
5 RadioButton CLIK 5

2.2.1.4

Button RadioButton disableF
ocus Toggle

label	Button
visible	false
disabled	true
disableConstraints	Button true t extFiel disa bled textFiel textFiel /e
selected	true RadioButton
data	RadioB utton
group	ButtonGroup RadioButton ButtonGroup *_root* *_root*

2.2.1.5

Object

- ***type:*** & 2«Ä
- ***target:*** & × Ð 7 Ä

RadioButton

show			true		
hide			false		
focusIn					
focusOut					
select	*Selected*		true		
stateChange	*State*	String	up	over	down
	CLIK				

rollOver		
	mouseIndex	-
	0-3	
rollOut		
	mouseIndex	
	0-3	
press		
	mouseIndex	
	0-3	
doubleClick	*doubleClickEnabled*	
	mouseIndex	
	0-3	
click		
	mouseIndex	-
	0-3	
dragOver		
	mouseIndex	-
	0-3	
dragOut		
	mouseIndex	-
	0-3	
releaseOutside		
	mouseIndex	-
	0-3	

RadioButton

```
myRadio.addEventListener("select", this, "onRadioToggle");
function onRadioToggle(event:Object) {
      // ⍁⍁
}
```

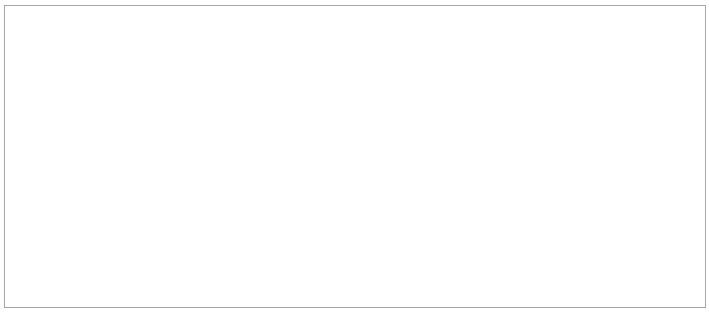
2.2.1.6 ⍁⍁

5 RadioButton

```
import gfx.controls.ButtonGroup;
var myGroup:ButtonGroup = new ButtonGroup("groupName");
radio1.group = myGroup;
radio2.group = myGroup;
```

```
        someMovie.radio1.group = myGroup;
        someMovie.radio2.group = myGroup;
```

2.2.2 ButtonGroup



. 21: ß ButtonGroup.

CLIK ButtonGroup
(gfx.controls.ButtonGroup)ÿ

ß

þ CLIKÿ CheckBoxÿ RadioButtonÿ ButtonGroup
ÿ

2.2.2.1 ÿ 交互

ÿ RadioButtonÿ
ÿ

2.2.2.2 ÿ 件ÿ 置

ÿ CLIKÿ ButtonGroupÿ MovieClipÿ

2.2.2.3 ÿ

ÿ ButtonGroupÿ

2.2.2.4 ÿ 属性

ÿ ButtonGroupÿ

2.2.2.5 事件

Object

- ***type:*** &
- ***target:*** &

ButtonGroup

change
• *item* : CLIK Button • *data* AS2
itemClick
• *item* : CLIK Button

ButtonGroup

```
myGroup.addEventListener("change", this, "onNewSelection");
function onNewSelection(event:Object) {
      if (event.item == myRadio) {
            // 执行操作
      }
}
```

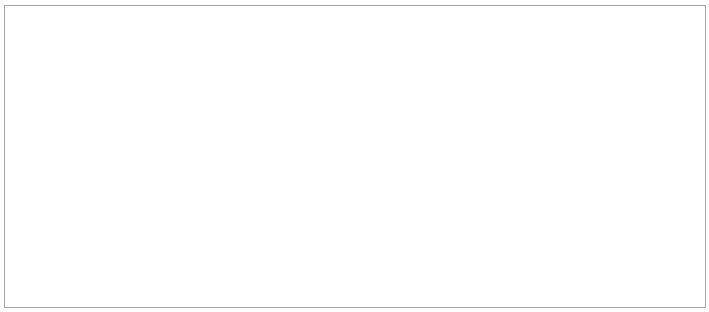
2.2.2.6 捕获方

```
var selectedRadio:Button = group.selectedButton;
```

ButtonGroup

```
radioBtn1.group.addEventListener("itemClick", this, "onClick");
```

2.2.3 ButtonBar



. 22: ButtonBar.

ButtonBar (gfx.controls.ButtonBar) ButtonGroup dataProvider dataProvider tab? UI

```
buttonBar.dataProvider = ["item1", "item2", "item3", "item4", "item5"];
```

2.2.3.1 交互

ButtonGroup

2.2.3.2 件 置

CLIK ButtonBar MovieClip

2.2.3.3 状

CLIK

2.2.3.4 属性

Flash IDE ButtonBar

visible		false	ButtonBar
disabled		true	buttonBar
itemRenderer			ID

direction	
spacing	*direction*
autoSize	true
buttonWidth	autoSize true

2.2.3.5 &

Object

- ***type:*** & 2«Ä
- ***target:*** & × Ð 7 Ä

ButtonBar

show	true
hide	false
focusIn	
focusOut	
change	• *index* 0 1 • *renderer* : CLIK Button • *item* : dataProvider AS2 Object • *data* dataProvider AS2 Object
itemClick	• *index* 0 1 • *item* : dataProvider AS2 Object • *data* dataProvider AS2 Object • *mouseIndex* 0-3

```
myBar.addEventListener("itemClick", this, "onItemClick");
function onItemClick(event:Object) {
	processData(event.data);
	// 执行操作
}
```

2.3

ScrollIndicator ScrollBar
Slider ScrollIndicator ScrollBar
Sl
ider

. 23: *Mercenaries 2*

2.3.1 ScrollIndicator

. 24: **ScrollIndicator**.

CLIK ScrollIndicator
(gfx.controls.ScrollIndicator) textField
TextArea
ID

2.3.1.1 用户交互

2.3.1.2 创建组件实例

CLIK 提供了 ScrollIndicator 组件的 MovieClip 实例。

- ***thumb***: CLIK 提供了 Button2 组件的实例。
- ***track***：CLIK 提供了 MovieClip2 组件的实例。

2.3.1.3 状态

. 25: ScrollIndicator 组件

2.3.1.4 可检查属性

scrollTarget		TextArea	textField
		textField	ScrollIndicator
visible		false	
disabled		false	
offsetTop			
offsetBottom			

2.3.1.5 事件

Object

- ***type:*** 事件 2
- ***target:*** 事件 7

ScrollIndicator

show	件可　属性在运行　已　置　true
hide	件可　属性在运行　已　置　false
scroll	• *position*

```
mySI.addEventListener("scroll", this, "onTextScroll");
function onTextScroll(event:Object) {
       trace("scroll position: " + event.position);
      // 执行操作
}
```

2.3.1.6 ¼

```
scrollInd.setScrollProperties(pageSize, minPos, maxPos, pageScrollSz);
scrollInd.positon = currPos;
```

A5□H□　　_rotation□□□K*4&&A5□j4&=□UA□

"dA□M0?□.,A5B□□

```
scrollInd.direction = "horizontal";
```

2.3.2 ScrollBar

. **26:** **ScrollBar.**

CLIK□　ScrollBar(gfx.controls.ScrollBar)

2.3.2.1 用户交互

ScrollBar

trackMode trackMode

(Shift)

2.3.2.2 内部组件和布局

CLIK ScrollBar MovieClip

- ***upArrow***: CLIK
- ***downArrow***: CLIK
- ***thumb***: CLIK
- ***track***: CLIK

2.3.2.3 状态

图 27: 滚动条 ScrollBar 组件

SrollBar ScrollIndicator

2.3.2.4 可访问属性

scrollTarget		TextArea textField
trackMode	scrollToCursor	scrollPage
visible		false
disabled		false
offsetTop		
offsetBottom		

2.3.2.5 事件

Object

- ***type:*** & 2«Ä
- ***target:*** & × Ð 7 。

ScrollBar

show	Ä & Ø Ø	& " Â 5 j true
hide	Ä & Ø Ø	& " Â 5 j false
scroll	 • *position*	

2.3.2.6 操作方法

```
mySB.addEventListener("scroll", this, "onListScroll");
function onListScroll(event:Object) {
	trace("scroll position: " + event.position);
	// 执行操作
}
```

```
scrollBar.setScrollProperties(pageSize, minPos, maxPos, pageScrollSz);
scrollBar.positon = currPos;
```

_rotation

```
scrollBar.direction = "horizontal";
```

scrollPage

```
scrollBar.trackScrollPageSize = pageSize;
```

2.3.3 Slider

. 28: Slider.

Slider(gfx.controls.Slider)

2.3.3.1 交互

home¼end

2.3.3.2 件

CLIK Slider MovieClip

Ö

- ***thumb***Ö CLIK
- ***Track***Ö CLIK

2.3.3.3

图 29: Slider

2.3.3.4 性

Slider

visible		false
disabled		false

value	Slider		
minimum			
maximum			
snapping	true		
snapInterval		false	
liveDragging	true	Slider	
[illegible]			
offsetRight			

2.3.3.5 事件

[illegible] Object[illegible]

- ***type:*** & 2«Ä
- ***target:*** & × Þ 7 Ä

Slider

show		true
hide		false
focusIn		
focusOut		
change	Slider	

Slider

```
mySlider.addEventListener("change", this, "onValueChange");
function onValueChange(event:Object) {
	trace("slider value: " + event.target.value);
	// 执行操作
}
```

2.4 [illegible]

CLIK NumericStepper OptionStepper ScrollingList TileList DropdownMenu NumericStepper DataProvider ListItemRenderer ScrollingList TileList

NumericStepper OptionStepper ScrollingList TileList ScrollIndicator ScrollBar DropdownMenu ScrollingList TileList

图 30: ***Mercenaries 2***

图 31: ***Halo Wars*** **'Difficulty Settings'**

图 32: *Mass Effect* 'Display Setting'

2.4.1 NumericStepper

图 33: Numeric Stepper.

NumericStepper
(gfx.controls.NumericStepper)

2.4.1.1 交互

NumericStepper

(Home)
(End)

2.4.1.2 件

NumericStepper MovieClip

- ***textField***: TextField
- ***nextBtn***: CLIK Button
- ***prevBtn***: CLIK Button

2.4.1.3

NumericStepper

- ***default*** enabled

- ***focused*** textField
- ***disabled***

. 34: NumericStepper

2.4.1.4 属性

MovieClip

visible		false
disabled		false
value	NumericStepper.	
minimum	NumericStepper	
maximum	NumericStepper	

2.4.1.5

Object

- ***type:*** 2
- ***target:*** 7

NumericStepper

show	件可 属性在运行 已 置 true
hide	件可 属性在运行 已 置 false
change	NumericStepper
stateChange	NumericStepper • *state* String “default” “focused” “disabled”

NumericStepper

```
myNS.addEventListener("change", this, "onValueChange");
```

```
	function onValueChange(event:Object) {
		trace("ns value: " + event.target.value);
		// 执行操作
	}
```

2.4.1.6 [illegible]

[illegible]

```
	ns.stepSize = 0.5;
```

□ □≥50□

```
	ns.labelFunction = function(value:Number) {
		switch(value) {
			case 1:
				return value + "st";
			default:
				return value + "th";
		}
	}
```

2.4.2 OptionStepper

图 35: [illegible] OptionStepper.

OptionStepper
(gfx.controls.OptionStepper),> NumericStepper[illegible]
dataProvider[illegible] [illegible] OptionStepper[illegible]
[illegible] [illegible] dataProvider[illegible]
[illegible]

```
	optionStepper.dataProvider = ["item1", "item2", "item3", "item4"];
```

2.4.2.1 用户交互

> NumericStepper组件和 OptionStepper组件的用户交互
方式基本一致，
均支持鼠标点击、键盘上下箭头、Page Up/Page Down、Home键和End键操作，
并且都可以通过dataProvider属性来设置数据源。

2.4.2.2 创建组件实例

下面以 CLIK NumericStepper组件为例，介绍如何在MovieClip面板中创建组件实例。

- ***textField***：TextField类型，
- ***nextBtn***：CLIK库中的 Button类型，用于增加数据源的索引值。
- ***prevBtn***：CLIK 库中的 Button类型，用于减小数据源的索引值。

2.4.2.3 状态

OptionStepper组件有以下几种状态：

- ***default***：默认状态，enabled状态。
- ***focused***：获得焦点状态，此时可以响应键盘事件。
- ***disabled***：禁用状态。

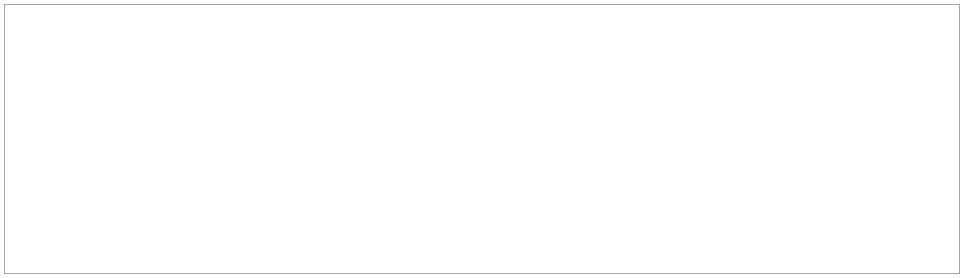


图 36: OptionStepper组件

2.4.2.4 设置属性

可以在 Component Inspector面板中为MovieClip设置以下属性：

visible	布尔值	false：隐藏
disabled	布尔值	false：禁用

2.4.2.5 事件

当用户与组件交互时，会触发Object事件：

- ***type:*** & 2«Ä
- ***target:*** & × Þ 7 Ä

OptionStepper

show		true
hide		false
change	OptionStepper	
stateChange	OptionStepper • *state* String “default” “focused” “disabled”	

OptionStepper

```
myOS.addEventListener("change", this, "onValueChange");
function onValueChange(event:Object) {
      trace("os value: " + event.target.selectedItem);
      // 执行操作
}
```

2.4.3 ListItemRenderer

. 37:Þ **ListItemRenderer.**

 ListItemRenderer (gfx.controls.ListItemRenderer)þ CLIKÞ Button

 ScrollingList TitleList Dropdownmenu

2.4.3.1 用户交互

默认情况下，ListItemRenderer会将Button组件作为其子显示对象。当用户与ListItemRenderer交互时，ListItemRenderer会将事件冒泡给其父级。

2.4.3.2 创建组件布置

要创建CLIK ListItemRenderer，请将MovieClip作为其基础。

- ***textField***（可选）TextField 实例，用于显示项目的文本。
- ***focusIndicator***：（可选）MovieClip实例，用于显示项目的焦点状态。MovieClip应该有两个帧标签：show和hide。

2.4.3.4 ð 属性

ListItemRenderer

label	ListItemRenderer	
visible		false
disabled		true

2.4.3.5 &

Object

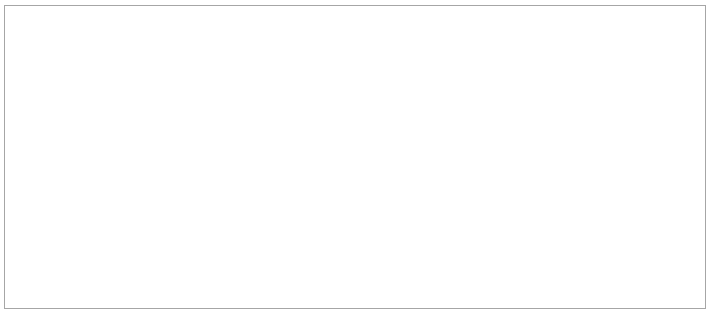
- ***type:*** & 2«Ä
- ***target:*** & × Ð 7 Ä

ListItemRenderer

show	true
hide	false
focusIn	
focusOut	
select	• *Selected* true
stateChange	按 状 已改 State：按 新状 。String 型。 “up”、“over”、“down”等。参考 CLIK 按 入 文档 取 的状 列表信息。
rollOver	• *mouseIndex* - 0-3
rollOut	• *mouseIndex* 0-3
press	• *mouseIndex* 0-3

	0-3	
click		
• *mouseIndex*	0-3	-
dragOver		
• *mouseIndex*	0-3	-
dragOut		
• *mouseIndex*	0-3	-
releaseOutside		
• *mouseIndex*	0-3	-

2.4.4 ScrollingList



. 39: **ScrollingList.**

ScrollingList
(gfx.controls.ScrollingList)
ScrollIndicator
ScrollBar dataProvider

```
scrollingList.dataProvider = ["item1", "item2", "item3", "item4"];
```

crollingList ListItemRenderer

FLA

2.4.4.1 用 交互

ScrollBar ScrollingList

ScrollingList

ScrollingList, *wrapping* *wrapping* nor
mal *wrapping* wrap
wrapping stick

(Page Up) ¼ (Page
down) (Home) ¼
(End) ScrollBar
ScrollingList ScrollaBar

D-
Pad CLIK; UI

2.4.4.2 件 置

ScrollingList ScrollingList

2.4.4.3 状

ScrollingList

- ***default*** F enable
- ***focused***
- ***disabled*** e

. **40: ScrollingList**

2.4.4.4 ð 属性

ScrollingList MovieClip

visible	false :
disabled	true
itemRenderer	ListItemRenderer *rendererInstanceName*
renderInstanceName	ScrollingList r r1 r2 r3 1
scrollBar	ScrollBar ScrollingList ScrollingList
margin	*rendererIns* *tanceName*
rowHeight	*rendererInstanceName*
paddingTop	*rendererInstanceName*
paddingBottom	*rendererInstanceName*
paddingLeft	*rendererInstanceName*
paddingRight	*rendererInstanceName*
thumbOffsetTop	
thumbOffsetBottom	
thumbSizeFactor	1.0 1.0

2.4.4.5 事件

Object

- ***type:*** 2
- ***target:*** 7

ScrollingList

show	true
hide	false
focusIn	
focusOut	
change	• *index* 0 1.
itemPress	• *renderer* CLIK • *item* dataProvider AS2 • *index* dataProvider 0 1 • *mouseIndex* 0 3
itemClick	• *renderer* CLIK • *item* dataProvider AS2 • *index* dataProvider 0 1 • *mouseIndex* 0 3
itemDoubleClick	• *renderer* CLIK • *item* dataProvider AS2 • *index* dataProvider 0 1 • *mouseIndex*

	0 3
itemRollOver	• *renderer* CLIK • *item* dataProvider AS2 • *index* dataProvider 0 1 • *mouseIndex* 0 3
[illegible]	• *renderer* CLIK • *item* dataProvider AS2 • *index* dataProvider 0 1 • *mouseIndex* 0 3

```
myList.addEventListener("itemClick", this, "onItemClicked");
function onItemClicked(event:Object) {
	trace("list item was clicked: " + event.renderer);
	// 执行操作
}
```

2.4.4.6 [illegible]

[illegible]

```
// ScrollingList[illegible] 'label'
// [illegible]
list.dataProvider = [{label: "one", data:1}, {label: "two", data:2}];

// [illegible]
// [illegible]
list.labelField = "name";
list.dataProvider = [{name: "one", data:1}, {name: "two", data:2}];

// [illegible]
```

```
    //□A5□ labelFunction□
list.labelFunction = function(itemObj:Object):String {
    // FkE□□O□□
}
list.dataProvider = [{p1: "foo", p2: 1}, {p1: "bar", p2: 2}];
```

2.4.5 TileList

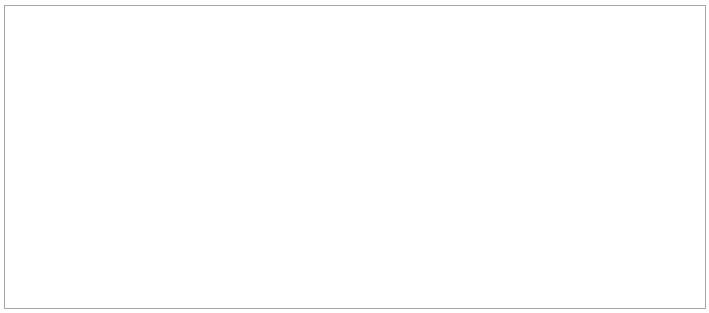


图 41: □ TileList.

□ □ TileList (gfx.controls.TileList)□ ScrollingList□ □ ; □. □ ScrollIndicator □ ScrollBar□ □ TileList□ ScrollingList□ □ □ □

```
tileList.dataProvider = ["item1", "item2", "item3", "item4", "item5"];
```

UA□ □ G>□ TileList□G>□□ ListItemRenderer4□□UG>□□□ FL A□□UMG>□□□iE04□?□?□□

2.4.5.1 用 交互

□ ScrollBar□ □ TileList□ □ □ TileList□ □ □

(Page Up) (Page down (Home)

(End) ScrollBar

TileList ScrollBar

2.4.5.2 件 置

TileList TileList

2.4.5.3

TileList

- ***default*** F enable
- ***focused*** e
- ***disabled*** e

. 42: **TileList**

2.4.5.4 ð 属性

TileList MovieClip

Visible		false
Disabled		true
itemRenderer		ListItemRenderer
		¶ *rendererInstanceName*
renderInstanceName		TileList r r1 r2 r3ÿ

	…..	1
scrollbar	ScrollBar TileList TileList	
margin	*stanceName*	*rendererIn*
rowHeight		*rendererInstanceName*
columnWidth		*rendererInstanceName*
externalColumnCount	*rendererInstanceName* TileList	
Direction		

2.4.5.5 &

Object

- ***type:*** & 2«
- ***target:*** & × 7 Ä

  TileList

itemClick	• *renderer* CLIK • *item* dataProvider AS2 • *index* dataProvider 0 1 • *mouseIndex* 0 3
itemDoubleClick	• *renderer* CLIK • *item* dataProvider AS2 • *index* dataProvider 0 1 • *mouseIndex* 0 3
itemRollOver	• *renderer* CLIK • *item* dataProvider AS2 • *index* dataProvider 0 1 • *mouseIndex* 0 3
[illegible]	• *renderer* CLIK • *item* dataProvider AS2 • *index* dataProvider 0 1 • *mouseIndex* 0 3

TileList

```
myList.addEventListener("focusIn", this, "onListFocused");
function onListFocused(event:Object) {
      trace("tile list was focused!");
```

```
            // 执行操作
        }
```

2.4.5.6 [illegible]

[illegible]

```
        // ScrollingList[illegible]            'label'
        //[illegible]
        list.dataProvider = [{label: "one", data:1}, {label: "two", data:2}];

        // [illegible]

        list.labelField = "name";
        list.dataProvider = [{name: "one", data:1}, {name: "two", data:2}];

        // [illegible]
        // FA5[illegible] labelFunction[illegible]
    list.labelFunction = function(itemObj:Object):String {
        // Fk[illegible]上构建一个    Logic to construct a label
    }
    list.dataProvider = [{p1: "foo", p2: 1}, {p1: "bar", p2: 2}];
```

2.4.6 DropdownMenu

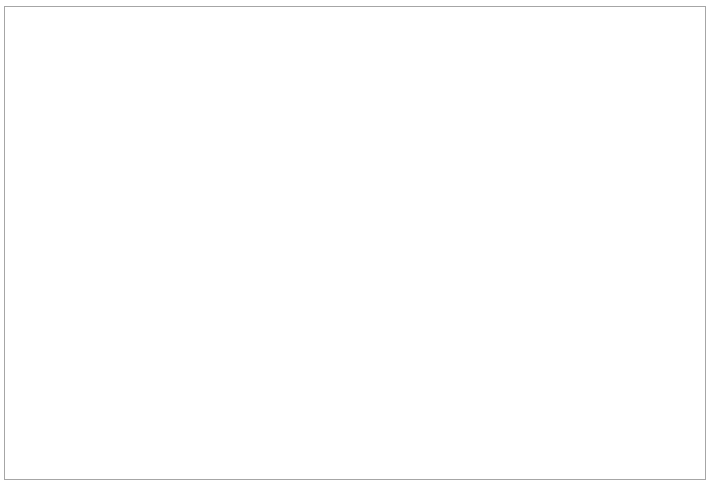


图 43: [illegible] DropdownMenu.

[illegible] DropdownMenu (gfx.controls.DropdownMenu)[illegible] Dr

opdownMenu

ScrollingList TileList ScrollBar ScrollIndicator dataProvider DropdownMenu dataProvide dataProvide

```
dropdownMenu.dataProvider = ["item1", "item2", "item3", "item4"];
```

UA DropdownMenu ListItemRenderer4 Dro pdownMenu ListItemRenderer FLA 63

M0UA DropdownMenu 0 ScrollBar /j ScrollIndicator 1Lt DropdownMenu s3P

2.4.6.1 交互

DropdownMenu (Enter) ScrollingList TileList ScrollBar DropdownMenu

2.4.6.2 件

DropdownMenu MovieClip

- ***textField*** TextField
- ***focusIndicator***: MovieClip hide

2.4.6.3

DropdownMenu ToggleButton CheckBox

- ***up***
- ***over***
- ***down***

- ***disabled*** (
- ***selected_up*** (
- ***selected_over*** (
- ***selected_ down*** (
- ***selected_disabled*** (

图 44: ; ù. **DropdownMenu**

DropdownMenu Button

DropdownMenu CLIK

2.4.6.4 ð 属性

ù. DropdownMenu

Visible		false
Disabled		false
Dropdown		ScrollingList **F** TileList DropdownMenu
dropdownWidth		-1
itemRenderer		
scrollbar		
Margin		
paddingTop		
paddingBottom		
paddingLeft		
paddingRight		
thumbOffsetTop		
thumbOffsetBottom		
thumbSizeFactor	thumb	1.0 thumb

	1.0 thumb
offset	
offset	
Extent	offsetX dropdownWidth - 1
Direction	

2.4.6.5 事件

Object

- ***type:*** 2
- ***target:*** 7

DropdownMenu *change* Button

show	true
Hide	false
focusIn	
focusOut	
change	• *index* 0 1. • *data :* AS2
select	• *Selected*
stateChange	按 状 已改 State：按 新状 。String 型。 “up”、“over”、“down”等。参考 CLIK 按 入 文档 取 的状 列表信息。
rollOver	• *mouseIndex* 0-3
rollOut	• *mouseIndex* 0-3
press	

	• *mouseIndex* 0-3
doubleClick	*doubleClickEnabled* true • *mouseIndex* 0-3
click	• *mouseIndex* 0-3
dragOver	• *mouseIndex* 0-3
dragOut	• *mouseIndex* 0-3
releaseOutside	• *mouseIndex* - 0-3

2.4.6.6 [illegible]

[illegible]

```
dropdown.addEventListener("click", this, "onClick");
function onClick(e:Object) {
   if (e.target.isOpen) {
      //□ open&□|□
   }
}
```

□□|□□□

```
    //□□tG>□x□/jG>□GG□M0?□□&□
  onEnterFrame = function() {
      dd.dropdown.scrollBar = "ScrollBar";
      onEnterFrame = null;
    }
```

□/j□E□r□B□□,□□

```
    // ScrollingList6□W=0          'label'
    // □=□B□□
    list.dataProvider = [{label: "one", data:1}, {label: "two", data:2}];

    // v_□□=□B□<,□□W□p/j□
    // □G>G}5□B□W□
    list.labelField = "name";
    list.dataProvider = [{name: "one", data:1}, {name: "two", data:2}];
    // □=□B□:□0□□FkE□:"□□rM0?□0□□
     //□A5□ labelFunction□
  list.labelFunction = function(itemObj:Object):String {
    // FkE□:□0□□
  }
  list.dataProvider = [{p1: "foo", p2: 1}, {p1: "bar", p2: 2}];
```

2.5

The Scaleform CLIK StatusIndicator ProgressBar StatusIndicator Progress Bar

图 45: *Mercenaries 2* Faction

2.5.1 StatusIndicator

图 46: **StatusIndicator.**

StatusIndicator component (gfx.controls.StatusIndicator)

2.5.1.1 交互

StatusIndicator

2.5.1.2 件

CLIK StatusIndicator MovieClip stop() gotoAndStop()

2.5.1.3

StatusIndicator

2.5.1.4 属性

StatusIndicator MovieClip

visible		false
disabled		false
value		
minimum		
maximum		

2.5.1.5

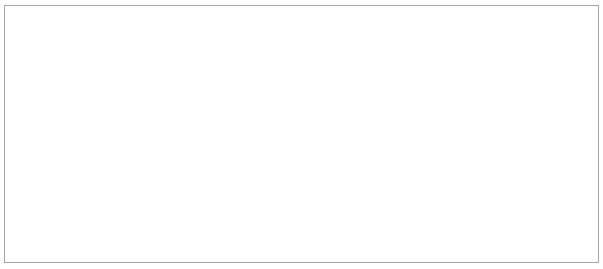
Object

- ***type:*** 2
- ***target:*** × 7

StatusIndicator

show	件可	属性在运行	已	置	true	
hide	件可	属性在运行	已	置	false	

2.5.2 ProgressBar



. 47: þ ProgressBar.

Ë ProgressBar (gfx.controls.ProgressBar)þ StatusIndicatorþ b ytesLoaded ¼ bytesTotalþ

2.5.2.1 用 交互

Ë ProgressBarþ.

2.5.2.2 þ 件 置

þ StatusIndicatorþ CLIKË StatusIndicatorþ MovieClipþ þ ProgressBarþ stop()þ Ö ProgressBarþ gotoAndStop()Ä

2.5.2.3 状

Ë ProgressBarþ

2.5.2.4 ð 属性

þ ProgressBarþ MovieClipþ

visible	þ falseþ
disabled	þ falseþ
target	Ë ProgressBarþ bytesLoaded ¼ bytesTotalI
mode	properties.Ë ProgressBarþ manual þ setProgressþ polled þ bytesLoaded

	bytesTotal	event	progress	byt
	esLoaded	bytesTotal		

2.5.2.5 [illegible]

[illegible] Object[illegible]

- ***type:*** [illegible] 2«[illegible]
- ***target:*** [illegible] × [illegible] 7 [illegible]

ProgressBar

show		true
hide		false
progress	ProgressBar	
complete		

```
myProgress.addEventListener("progress", this, "onProgress");
myProgress.addEventListener("complete", this, "onProgress");
function onProgress(event:Object) {
	if (event.type == "progress") {
		// [illegible]
	} else
	{
		trace("Loading complete!");
	}
}
```

2.6

Scaleform
CLIK UI Dialog UILoader
¼
ViewStack Dialog UI UILoader
ViewStack

. 48: *Halo Wars* Dialog

2.6.1 Dialog

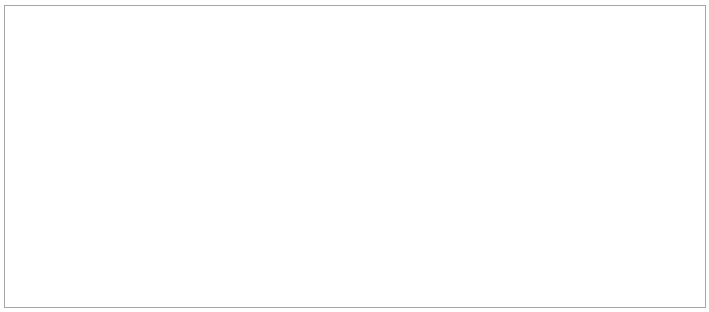


图 49: Dialog

CLIK Dialog
(gfx.controls.Dialog)
MovieClip MovieCli
p Dialog.show()

2.6.1.1 用 交互

Dialog

2.6.1.2 件 置

CLIK Dialog MovieClip

- ***closeBtn*** ( ) CLIK Button2
- ***cancelBtn*** ( ) CLIK Button2 # cancel
- ***submitBtn*** ( ) CLIK Button2 OK
- ***dragBar*** ( ) MovieClip 2

2.6.1.3 状

Dialog MovieClip

2.6.1.4 属性

Dialog MovieClip

visible		false	&
disabled		false	

2.6.1.5 事件

Object

- ***type:*** & 2
- ***target:*** & × 7

Dialog

show	件可 属性在运行 已 置 true
hide	件可 属性在运行 已 置 false
submit	Dialog • *data* Dialog getSubmitData true getSubmi

	tData AS2
close	Dialog

```
myDialog.addEventListener("submit", this, "onLoginSubmit");
function onLoginSubmit(event:Object) {
        trace("Received data from dialog: " + event.data);
}
```

2.6.2 UILoader

CLIK UILoader
(gfx.controls.UILoader)

SWF/GFX UILoaders

Scaleform

2.6.2.1 交互

UILoader SWF/GFX UILoader

2.6.2.2 件

CLIK UILoader MovieClip

- ***bg*** ( ) MovieClip
  2 UILoader

2.6.2.3

UILoader SWF/GFX UILoader

2.6.2.4 属性

UI UILoader MovieClip

Visible	false	
autoSize	true UILoader	
maintainAspectRatio	true UILoader	false UILoader
Source	SWF/GFX	
Timeout		UILoad er ioError

2.6.2.5 事件

Object

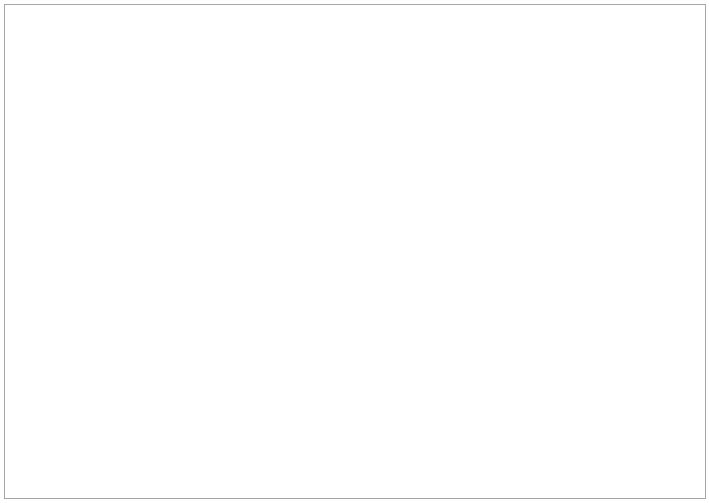
- ***type:*** 2«Ä
- ***target:*** × Ð 7 Ä

Ulloader

show	件可 属性在运行& 已 置j	true
hide	件可 属性在运行& 已 置j	false
progress		a b
loaded	0¼100	
complete	¼	
ioError		

```
myUILoader.addEventListener("ioError", this, "onLoadingError");
function onLoadingError(event:Object) {
      displayErrorMessage("Could not load" + event.target.source);
}
```

2.6.3 ViewStack



. 50: ViewStack

CLIK ViewStack
(gfx.controls.ViewStack) TabBo
x
Accordion ViewStack
RadioButton

2.6.3.1 用 交互

ViewStack ViewStack

2.6.3.2 件 置

MovieClip CLIK ViewStack

2.6.3.3 状

ViewStack ViewStack

2.6.3.4 属性

ViewStack MovieClip

visible	false
cache	true ViewStack targetGroup
targetGroup	ButtonGroup change ID Radio Buttons Flash IDE ID

2.6.3.5 事

ViewStack

3

Scaleform
CLIK

3.1

Scaleform CLIK

3.1.1

CLIK Flash

1. Flash
2.
3. Flash
4.
5. 1px×
6. OKÄ

图 51: **Grid**

SWF 1 2 2
2

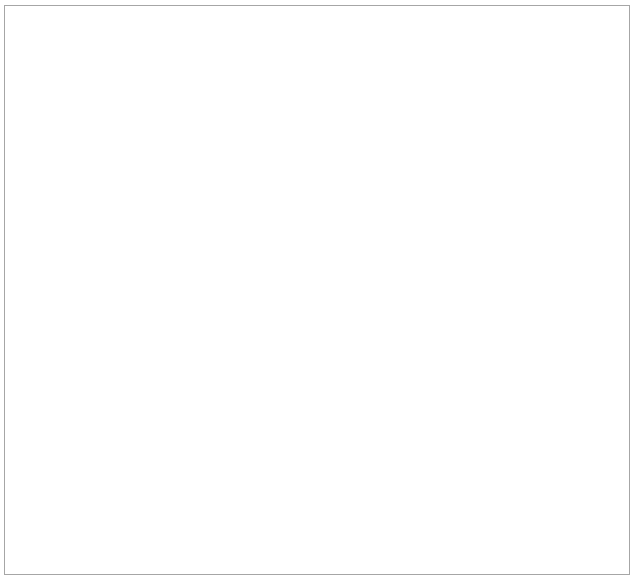


图 52:

3.1.2

X Flash

Scaleform

X Flash Photoshop® PNG

3.1.3

–

3.1.4

X Flash

3.1.5

3.1.6 2

2 2

- 16x16
- 32x32
- 64x64
- 128x128
- 128x256
- 256x256
- 512x256
- 512x512

3.2

Flash Scaleform CLIK

3.2.1

X Flash

3.2.1.1 1Å

FLA CLIK FLA

1. *CLIK_Components.fla*&Ä
2. FLA *Copy* FL A *Paste*Ä
3. FLA *Rename* þ Button
4. FLA *Properties*Ä
5. i *Identifier* 3Ä
6. *Paste*

3.2.1.2 [illegible] 2Å

[illegible]

1. [illegible] *library*[illegible] *Properties*Ä
2. X *Properties*[illegible] [illegible] gfx.controlsButton[illegible] *Class*[illegible] (CTRL+C)[illegible] [illegible]
3. [illegible] *Cancel*Ä
4. [illegible] *Duplicate*Ä
5. X *Duplicate Symbol* [illegible] *Export*¯ *ActionScript*[illegible]
6. [illegible] *Class*j [illegible] (CTRL+V)[illegible] textFieldÄ
7. [illegible] OKÄ
8. [illegible] *library*[illegible] *Component Definition*Ä
9. [illegible] *Class* textField[illegible] (CTRL+V)[illegible] textFieldÄ
10. [illegible] OKÄ

图 53: [illegible]

图 **54:**

3.3

Scaleform
CLIK

CLIK

CLIK Flash FLA

- X Flash
- X Flash
- ; Photoshop F Illustrator® FlashÄ

CLIK

3.3.1 StatusIndicator

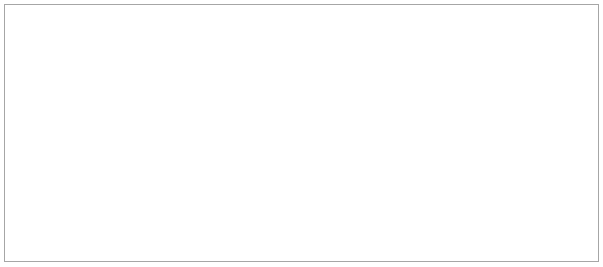


图 55: **StatusIndicator.**

StatusIndicator

indicator

track *Track* *Indicator*

Photoshop

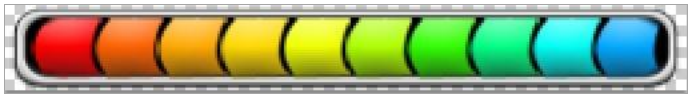


图56: **StatusIndicator PSD**

图 57　Photoshop　StatusIndicator PSD

1. X Photoshop　　StatusIndicator
2. 　　PSD
3. X Flash　　*File*　　*Import -> Import to Stage*
4. 　　StatusIndicator　　PSD
5. X *Import*　　*Convert layers to:*　　Layers
6. 　　OK
7. 　　Flash　　PSD
   　PSD　　1`　10È 1　　10
   　　layer 1
8.
9.
10. X *layer 1*　　6
11. 　　*Insert*　　*layer 1.*　　11
12. X *layer 2.*　　(Ctrl+X)
13. 　　*layer 1.*　　11　　(Ctrl+Shift+V)　　*layer 1.*
14. 　　10　　(*layer 1)*　　5　　*layer 2*　　11　　*layer 3*　　*16*　　*layer 4*　　21　　10Z PSD.　　51
15. 　　2-10

16.
    Remove Frames 55
17. *indicator layer*
18. *track* . *track.*
19. *layer 1* indicator
20. *indicator.* *track.* *actions layer.* *track.* *indicator.* ×

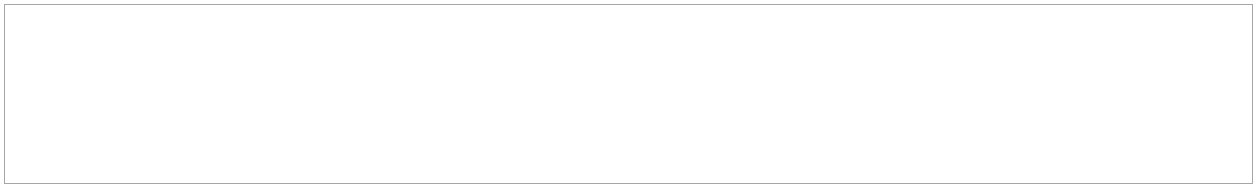


图 **58:**

21. StatusIndicator
22. 1` 10.
23. &
24. 3

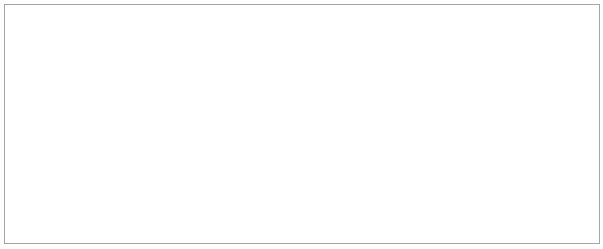


图**59: StatusIndicator**

3.4

Scaleform CLIK

3.4.1

Scaleform SWF Scaleform
Flash UI

3.4.2

Flash Scaleform
Scaleform

Scaleform

Scaleform textFidld Slate Mobile Slate Mobile CJK ASCII Sla te Mobile CJK

textFields Scaleform 3.4.4 Scalefor m Scaleform UI

3.4.3 X textField

textFields textFidld (Window > Property *Embed* *OK* F LA FLA

3.4.4

Scaleform SWF SWF Scaleform

Scaleform textField gfxfontlib.fla swf Scaleform

Scaleform fontconfig.txt

fontconfig.txt Scaleform

$TITLE

$TITLE=Scaleform Scaleform

Scaleform Scaleform

4

 Scaleform CLIK

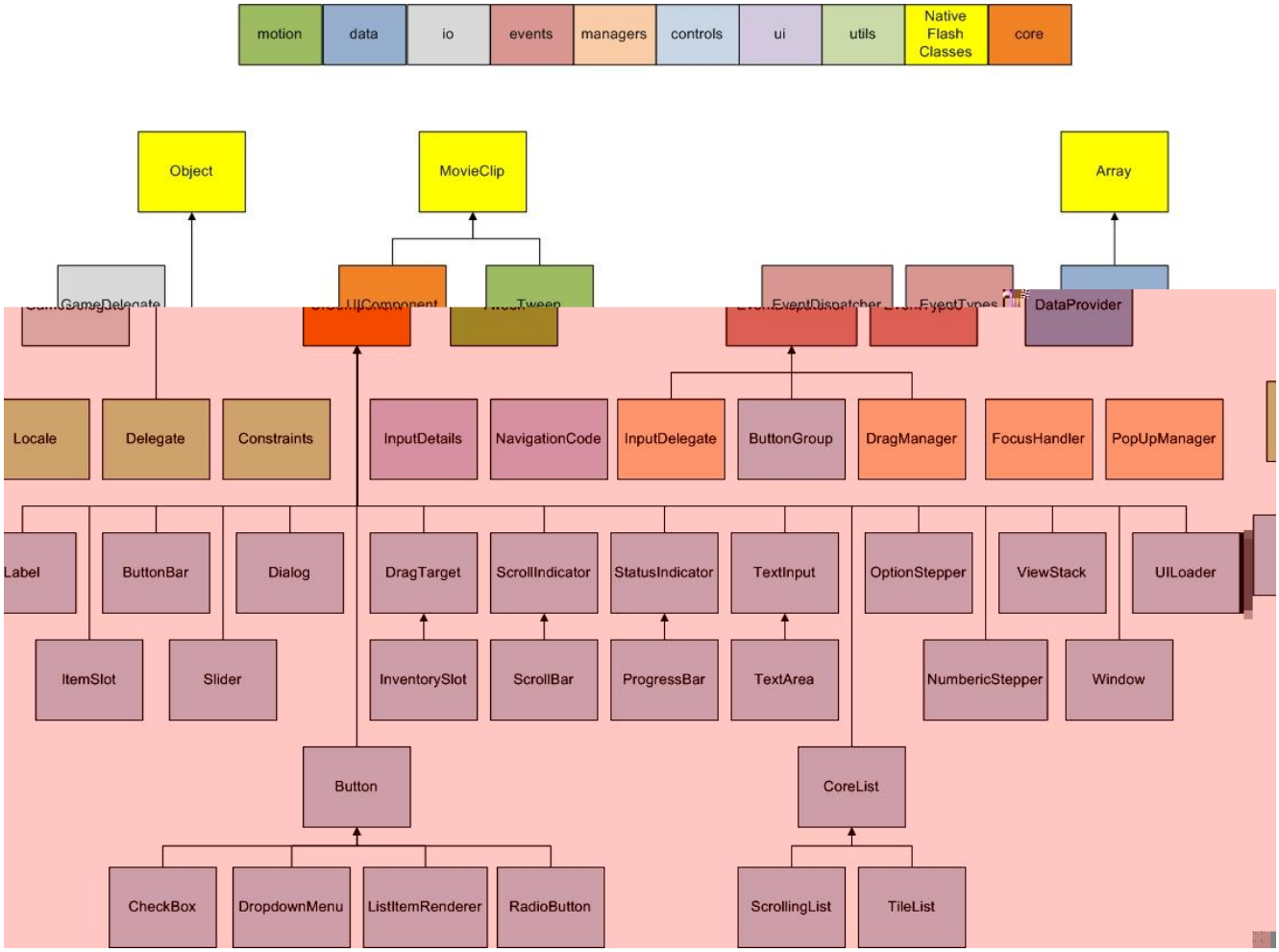


图 60: Scaleform CLIK

4.1 UI UIComponent

 Scaleform
CLIK UIComponent (gfx.core.UIComponent).
 Scaleform

UIComponent
MovieClip Flash MovieClip
/ UIComponent

- ***disabled***;
- ***visible***;
- ***focused***;
- ***width***;
- ***height***;
- ***displayFocus***: true

UIComponent
- ***configUI***:
- ***draw***:
- ***changeFocus***:
- ***scrollWheel***: 当鼠标光标在组件上方滚动滚轮时被调用。

UIComponent EventDispatcher

4.1.1

onLoad()
- MovieClip
- configUI()
- validateNow() draw()Ä

4.2

 Scaleform
CLIK

4.2.1

CLIK Button ListItemRenderer RadioButton CheckBox ScrollBar Button CLIK Button (gfx.controls.Button) Button

- *up*
- *over*
- *down*
- *disabled*

Button CLIK Button selected_

1

- *up*
- *over*
- *down*
- *disabled*
- *selected_up*
- *selected_over*
- *selected_ down*
- *selected_disabled*

CLIK CLIK

CLIK getStatePrefixes()

```
private function getStatePrefixes():Array {
      return (_selected) ? ["selected_",""] : [""];
}
```

CLIK selected_ getStatePrefixes()

stateMap

CLIK

```
private var stateMap:Object = {
        up:["up"],
        over:["over"],
        down:["down"],
        release: ["release", "over"],
        out:["out","up"],
        disabled:["disabled"],
        selecting: ["selecting", "over"],
        kb_selecting: ["kb_selecting", "up"],
        kb_release: ["kb_release", "out", "up"],
        kb_down: ["kb_down", "down"]
}
```

getStatePrefixes()

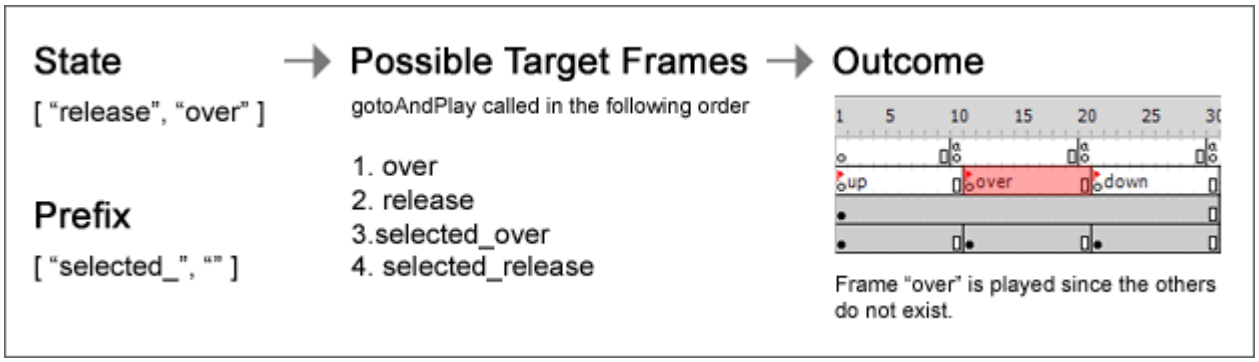


图 61:

CLIK Scrolli

ngListÃ OptionStepperÃ Slider

¼TextArea

- ***default***
- ***focused***
- ***disabled***

4.2.3

- ***default***
- ***disabled***

4.2.4

StatusIndicator ProgressBar

50

0 10 5 50% gotoAndStop() 25 50 50%

updateValue()

4.3

Scaleform CLIK *event*
mode

ActionScript 2

EventDispatcher (gfx.events.EventDispatcher) API CLIK
EventDispatcher.initialize() CLIK UIComponent
super() UIComponent EventDispatcher Eve
ntDispatcher

4.3.1

4.3.1.1

addEventListener()
removeEventListener()
String
=

EventTypes (gfx.events.EventTypes)
(EventTypes.SHOW show)

```
buttonInstance.addEventListener("itemClick", this, "callBack");
function callBack(eventObj:Object):Void {
	buttonInstance.removeEventListener("itemClick", this, "callBack");
}
```

在本例中按钮实例设置为监听“itemClick”事件并在接收到事件时执行函数“callback”。回收函数移除事件监听器。

事件对象的属性基于原始类型但各不相同。CLIK组件产生的详细事件对象（及其属性）列表，请参考内置组件相关章节。

4.3.1.2

CLIK dispatchEvent()

```
dispatchEvent({type:"itemClick", item:selectedItem});
```

4.4

Scaleform
CLIK
Scaleform A
ctionScript
] Selection.setFocus FocusHandler (gfx.managers.FocusHandler)
FocusHandler

4.4.1

CLIK
Tab true

```
Selection.setFocus(firstComponentOrMovieClip);
```

MovieClips onRelease
onRollOver Scaleform Flash
MovieClip focusEnabled tabEnabled

ScrollBar
FocusHandler

displayFocus tru
e Slider
UIComponent.changeFocus()

```
function changeFocus():Void {
	track.displayFocus = _focused;
}
```

[illegible]
[illegible] tabEnabled false

```
background.tabEnabled = false;
```

4.4.2 [illegible]

[illegible] ScrollingList OptionStepper
ButtonBar[illegible] Flash¼Scaleform
] Selection[illegible]

[illegible]

1. [illegible] tabEnabled = false [illegible] OptionStepper[illegible]
2. [illegible] focusEnabled = true
3. [illegible] displayFocus true changeFocus

[illegible]

4.5 [illegible]

[illegible] Scaleform CLIK[illegible] Flash 8
MovieClip[illegible]. MovieClip[illegible]
[illegible]. CLIK[illegible]

4.5.1 [illegible]

[illegible] InputDelegate[illegible] class
(gfx.managers.InputDelegate)[illegible] InputDelegate[illegible] InputDetails[illegible] (gfx.u
i.InputDetails)[illegible] InputDelegate.readInput()[illegible]
[illegible] InputDetails[illegible] InputDelegate[illegible]

InputDetails[illegible]

- 2[illegible] key
- [illegible]
- [illegible] 关键事件 由键的按下与弹起动作
  而产生 ，相应的输入细节的数值参数也分别为“keyUp”或“keyDown” ；
- navEquivalent(navigation equivalent)[illegible] “up”Ã “down”Ã “left” [illegible]
  “right”[illegible] NavigationCode 2«
  [illegible].NavigationCode[illegible]

FocusHandler[illegible] InputDelegate[illegible]
[illegible] the handleInput()[illegible]

[illegible] InputDetails¼pathToFocus[illegible]

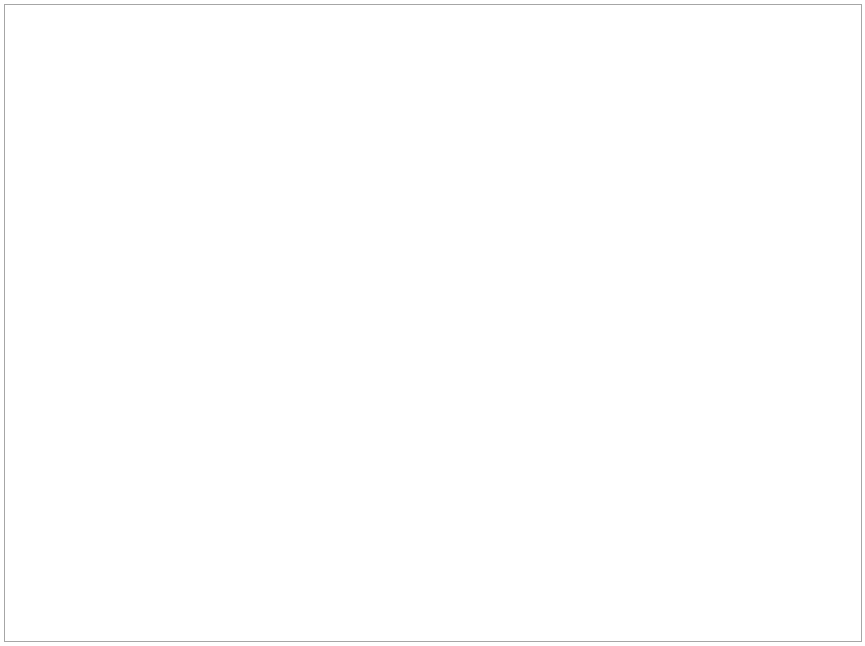


图 62: handleInput()

FocusHandler handleInput()
false null navEquivalent Scalef
orm

```
function handleInput(details:InputDetails, pathToFocus:Array):Boolean {
      if (details.navEquivalent == "left")
      {  // □  NavigationCode.LEFT
            doSomething();
            return true;
      }
      return false;
}
```

!4&4*6E6&PFB5□ pathToFocusG#,;04&F>4*6IF□ trueF fa
lse0=?A'V&D],;04&6!.>|E4*6IBBF>|.APFB0
3D1=EEPFB,I□

```
function handleInput(details:InputDetails, pathToFocus:Array):Boolean {
      var nextItem:MovieClip = pathToFocus.shift();
      var handled:Boolean = nextItem.handleInput(details, pathToFocus);
      if (handled) { return true; }
      //4*6.1
      return false; //4*6Ij      true
}
```

DropdownMenu handleInput

pathToFocus

```
function handleInput(details:InputDetails, pathToFocus:Array):Boolean {
      var handled:Boolean = dropdown.handleInput(details);
      if (handled) { return true; }
      // 
      return false; //  true
}
```

InputDelegate.instance

FocusHandler

```
InputDelegate.instance.addEventListener("input", this, "handleInput");
function handleInput(event:Object):Void {
      var details:InputDetails = event.details;
      if (details.value = Key.TAB) { doSomething(); }
}
```

InputDelegate InputDelegate

W A S D

Nintendo

Wii™ CLIK **Z.** SWF

```
myButton.addEventListener("click", this, "onCursorClick");
function onCursorClick(event:Object):Void {
      switch (event.mouseIndex) {
      …
      }
}
```

4.6

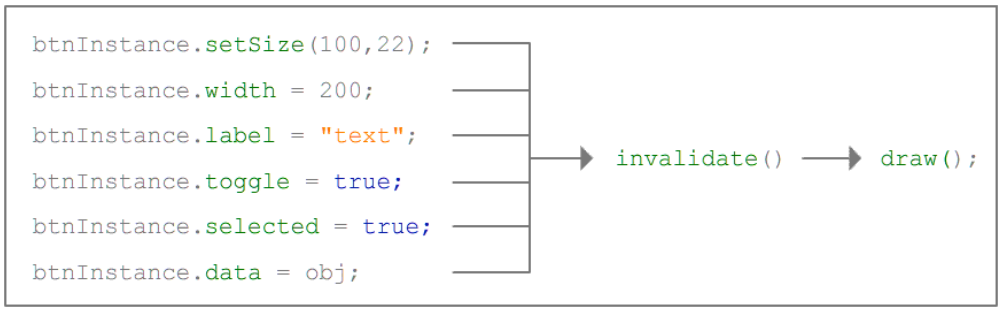


图 63: CLIK

4.6.1

UIComponent.invalidate()

UIComponent.draw() invalidate() draw()

UIComponent.validateNow()

4.7

Scaleform CLIK

1.

2. scale9grid

Flash
scale9Grid Flash/Flex
container

CLIK Scaleform scale9Grid
CLIK

container

Flash
IDE setSize()
Flash
IDE MovieClips LivePrevi
ews Scaleform
Player CLIK

4.7.1 Scale9Grid

Scaleform Player
Scale9Grid MovieClip scale9grid Flash MovieClips
scale9grid Flash
Player Scaleform

MovieClip scale9grid. Scale
9grid.

4.7.2

Constraints (gfx.utils.Constraints)
ScrollBar
Constraints

ScrollBar configUI()
draw()
draw()

```
    private function configUI():Void {
        …
        constraints = new Constraints(this);
        // A:AK   upArrow}
        constraints.addElements(downArrow, Constraints.BOTTOM);
        constraints.addElenent(track, Constraints.TOP|Constraints.BOTTOM);
        …
    }
```

```
    private function draw():Void {
          …
        constraints.update(__width, __height);
          …
    }
```

dataProvider dataProvider

Scaleform CLIK IDataProvider (gfx.interfaces.IDataProvider) API

dataProvider

dataProvider CoreList (ScrollingList, TileList) OptionStepper DropdownMenu CLIK IDataProvider API IDataProvider

4.8.1 最佳使用方法

DataProvider (gfx.data.DataProvider) initialize() dataProvider. ActionScript dataProvider dataProvider dataProvider IDataProvider

```
    myComponent.dataProvider = [ "data1",
                                 4.3,
                                       {label:"anObjectElement", value:6} ];
```

dataProvider

- *length*:
- *requestItemAt*: dataProvider. OptionStepper×
- *requestItemRange*: dataProvider ScrollingList×
- *indexOf*:
- *invalidate*: dataProvider dataChange dataProvider

dataProvider ActionScript 2
dataProvider ExternalInterface.call

4.9

Scaleform CLIK Tween (gfx.motion.Tween) Flash 8 Tween Scaleform Tween easing mx.transitions.easing.*

CLIK Tween Tween Flash 8 MovieClip MovieClips Tween CLIK CLIK Tween onEnterFrame MovieClip Tween Tween MovieClip Tween Tween Tween MovieClip Tweens

4.9.1

Tween MovieClip tweenTo() ¼ tweenFrom() tweenTo() MovieClip tweenFrom( )

Tween MovieClip Tween T ween.init()

```
import mx.transitions.easing.*;
import gfx.motion.Tween;
Tween.init();     // ⊠□ tween□    MovieClip□
// ⊠|0□   10J,□ Tween⊞k□⊟□5□          alpha□□,□□
myMovieClip.tweenTo(1, {_x: 200, _alpha: 0}, Strong.easeIn);
```

□ Tween⊞k5⊠>FJ.M0□ Tween□B□K*0□ onTweenComplete()=□

```
import mx.transitions.easing.*;
import gfx.motion.Tween;
Tween.init();
mc.onTweenComplete = function() {
      trace("Tween has finished!");
```

```
        }
        mc.tweenTo(5, {_rotation: 20},    Bounce.easeOut);
```

4.10

Scaleform
CLIK PopUpManager (gfx.managers.PopUpManager)
PopUpManager

4.10.1

PopUpManager createPopUp()
MovieClip CLIK ID
MovieClips

```
        import gfx.managers.PopUpManager;
        PopUpManager.createPopUp(context, "MyToolTipLinkageID",
                                          {_x: 200, _y:200}, relativeTo);
```

Scaleform topmostLevel
root SWF
topmostLevel

topmostLevel
true

4.11

DragManager
(gfx.managers.DragManager)

startDrag()
MovieClip ID
stopDrag()
DragManager> InputDelegate

4.11.1

图 64: DragDemo

DragManager DragManager Scaleform CLIK DragTarget (gfx.controls.DragTarget) DragTarget UIComponent CLIK DragManager

DragTarget DragTarget DragTarget DragManager dragBegin dragEnd DragTarget

CLIK DragTarget DragManager DragTarget InventorySlot ItemSlot CLIK

InventorySlot (gfx.controls.InventorySlot) DragTarget DragManager InventorySlot

ItemSlot
(gfx.controls.ItemSlot) InventorySlot
ItemSlot DragTarget CLIK ItemSlot

4.12

4.12.1 Delegate

Delegate (gfx.utils.Delegate) Scaleform
CLIK DragManager

```
eventProvider.onMouseMove = Delegate.create(this, doDrag);
```

4.12.2 Locale

Locale (gfx.utils.Locale)
Scaleform ActionScript
Button Label TextInput
Locale.getTranslatedString()
Locale

5 Î »

Scaleform CLIK

5.1 p

Scaleform CLIK

5.1.1 textField, ListItem Renderer

Michael	Jordan
Roger	Federer
Michael	Schumacher
Tiger	Woods
Babe	Ruth
Wayne	Gretzky

图 65: ScrollingList

ListItemRenderer ListItemRenderer. textField
textField
.

ListItemRenderer ListItemRenderer
ScrollingList textFields
. dataProvider Ö

```
      list.dataProvider = [{fname: "Michael", lname: "Jordan"},
                           {fname: "Roger",   lname: "Federer"},
                           {fname: "Michael", lname: "Schumacher"},
                           {fname: "Tiger",   lname: "Woods"},
                           {fname: "Babe",    lname: "Ruth"},
                           {fname: "Wayne",   lname: "Gretzky"},
                           {fname: "Usain",   lname: "Bolt"}];
```

5 ,) B W fname
lname B W 6 n / j G > N , 5 textField]

ListItemRenderer textField textField
ListItemRenderer MyItemRenderer ListIte
mRenderer *MyItemRenderer.as*
FLA

```
import gfx.controls.ListItemRenderer;

class MyItemRenderer extends ListItemRenderer {

	public var textField1:TextField;
	public var textField2:TextField;

	public function MyItemRenderer() { super(); }

	public function setData(data:Object):Void {
		this.data = data;
		textField1.text = data ? data.fname : "";
		textField2.text = data ? data.lname : "";
	}

	private function updateAfterStateChange():Void {
		textField1.text = data ? data.fname : "";
		textField2.text = data ? data.lname : "";
	}
}
```

ListItemRenderer , setData
 updateAfterStateChange setData (Scrollin
gList TileList ListItemRenderer textField, MyItemRender
er+ textField, updateAfterStateChange
 ListItemRenderer (, , textField
,

ListItemRenderer2, dataProvider
(Li
stItemRenderer textFild textField1
 textField2, MyItemRenderer MyItemRenderer4
, itemRenderer ListItemRenderer MyItemRenderer

FLA dataProvider fname
lname

5.1.2

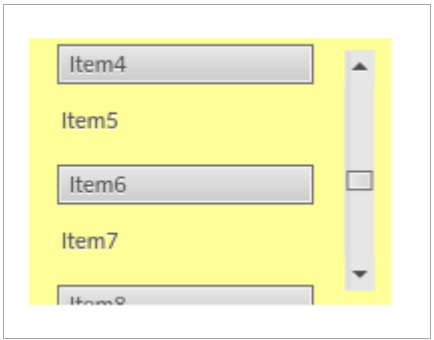


图 66: CLIK

CLIK ScrollBar.

- ***actions***: ActionScript.1
- ***scrollbar***: CLIK ScrollBar sb
- ***mask*** MovieClip
- ***content***
- ***background***

content y 0

ActionScript

.1 *code*.

```
// 133 ?
sb.setScrollProperties(1, 0, content._height - 133);
sb.position = 0;

sb.addEventListener("scroll", this, "onScroll");
function onScroll() {
	content._y = -sb.position;
}
```

setScrollProperties

Scaleform HUD Scaleform

5.2

 Scaleform
CLIK Flash¼ ActionScript 2Ä

5.2.1 X ScrollingList ; TreeView

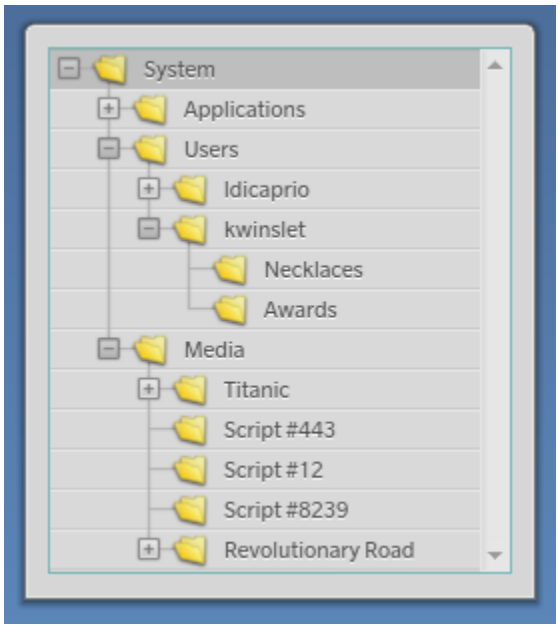


图 67: TreeView

 TreeView Demo
(*Resources/AS2/CLIK/demos/TreeViewDemo.fla*)X ScrollingList
 ListItemRenderer ¼ DataProviderÄ

 ScrollingList

-
-
  ListItemRenderer

- ListItemRenderer

ListItemRenderer Tre
eViewItemRenderer TreeViewDataProvider
TreeViewConstants *Resources/AS2/CLIK/demos/com/*...

1 ActionScript
2

```
var root:Object =
{
      label: "System",
      nodes:[
                {label: "Applications",
                 nodes:[
                            {label: "CGStudio",
                             nodes:[
                                        {label: "Data"},
                                        {label: "Samples"},
                                        {label: "Config"}
                                       ]
                               },
                            {label: "Money Manager"},
                            {label: "Role Call v6"},
                            {label: "Super Draft",
                             nodes:[
                                        {label: "Templates"},
 …
```

label
nodes ScrollingList

TreeViewDataProvider

ListItemRenderer /

TreeViewDataProvider CLIK

TreeViewItemRenderer ScrollingList TreeViewDataProvider
ListItemRenderer
TreeViewItemRenderer
tab ListItemRenderer tex
tField

TreeViewConstant

- open nodeÃ closed nodeÃ
-
-

ScrollingList ScrollBar
1 . / / Sc
rollingList Ä

5.2.2

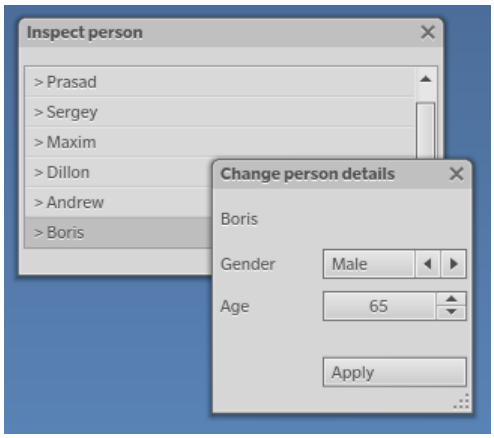


图 68:

Window Demo
(*Resources/AS2/CLIK/demos/WindowDemo.fla*) .
Resources/AS2/CLIK/demos/com/scaleform/ .

UIComponent

-
-
- ×
-
-

Window2

Title	
formPadding	
formType	symbol swf symbol swf SWF
formSource	*formType* symbol SWF& *formType* swf
allowResize	B
minWidth, maxWidth, minHeight, maxHeight	0È

.1

```
[Inspectable(name="formType", enumeration="symbol,swf"]
private var _formType:String = "symbol";
```

Flash _formTy

pe formType,

```
[Inspectable(defaultValue="Title")]
public function get title():String { return _title; }
public function set title(value:String):Void {
      _title = value;
```

```
            invalidate();
      }
```

CLIK

UIComponent configUI()
draw() configUI() draw()
B3+ configForm()

Flash 8
MovieClipLoader Scaleform MovieCli
pLoader.onLoadComplete() configForm()

draw()
validateNow()

UIComponent draw() draw()
validateNow() SWF
draw() validateNow()

```
      private function onMouseDown() {
            var targetObj:Object = Mouse.getTopMostEntity();
            while (targetObj != null && targetObj != _root)
            {
                  if (targetObj == this) {
                        swapDepths(_parent.getNextHighestDepth());
                        return;
                  }
            targetObj = targetObj._parent;
            }
      }
```

WindowManager zE Win
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